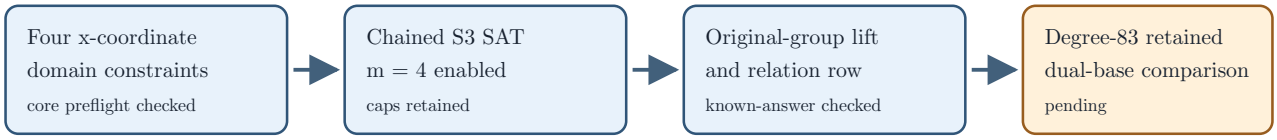
1. Four-summand chained-S3 entry point for retained factor bases

Status: Native solver integration and small-degree correctness validation. Degree-83 natural relation yield, useful rank, and runtime remain unmeasured.

The full-width chained-S3 circuit supports four summands, but the index-calculus decomposition and dispatch guards previously admitted only five or six. This change enables m = 4 while retaining finite-domain preflight, solver caps, original-field checks, group lifting, and relation-rank checks.

m = 4 retained-base relation path



Checked: small-degree exact group sums, signed-Frobenius domains, rank/answer path, cap handling.

Open: authenticated K=2048 and structured d12 imports, natural queries, exact rank, cold accounting.

Source: src/cryptanalysis/koblitz_index_calculus.rs; study n83_chained_s3_m4_20261017.

1.1. Checked properties

Release library tests exercise four-, five-, and six-summand witnesses on degree-7 known-answer fixtures. They replay sums in the original group and check signed-Frobenius and ordered-domain modes, incomplete-orbit rejection, exact rank, answer verification, and cap handling. Exact commands and statuses are in `VALIDATION.md`.

The degree-83 m = 4 construction preflight gives 63,578 maximum circuit variables. The 2,001-representative orbit domain adds 21,012 variables, 697,808 clauses, and 2,656 XOR rows. These are derived bounds, not measured solve resources.

1.2. Remaining work

The retained-base adapter must authenticate both the K = 2,048 explicit base and the 2,001-column structured base and run a common bounded m = 4 query corpus. Independent group replay, exact rank, complete cold costs, and matched references are required before comparing quality. The existing N83 prebuilt-base problem record remains open.

Source: `src/cryptanalysis/koblitz_index_calculus.rs`, `src/cryptanalysis/binary_semaev_chain_sat.rs`, and `research/n83_prebuilt_factor_base_20261016/REVIEW.md`.